

ECRA: An Evidence-Based Framework for Structured Course Review Analysis

Jaehee Jung¹, Dayoung Jeong², Kyungsik Han^{1, 2}

¹ Department of Data Science, Hanyang University, Seoul, Republic of Korea

² Department of Artificial Intelligence, Hanyang University, Seoul, Republic of Korea

Motivation

이 논문은 정부(교육부)의 재원으로 한국산업기술진흥원의 지원(P0030395, 첨단산업인재양성부트캠프)과 2026 정부(과학기술정보통신부)의 지원을 받아 수행된 연구임 (RS-2026-25490215)

Limitations & Research Gap

- Generative AI diversifies **students' learning experiences**
- Course reviews are **unstructured**, with **repetitive** and **conflicting opinions**
- Lack of systematic analysis for educational course reviews**

The Need for Evidence-Based Review Structuring

- Structured representations** preserve educational evidence throughout the analysis process
- Evidence-based review structuring** enables reliable hierarchical summarization and Learning Gap analysis

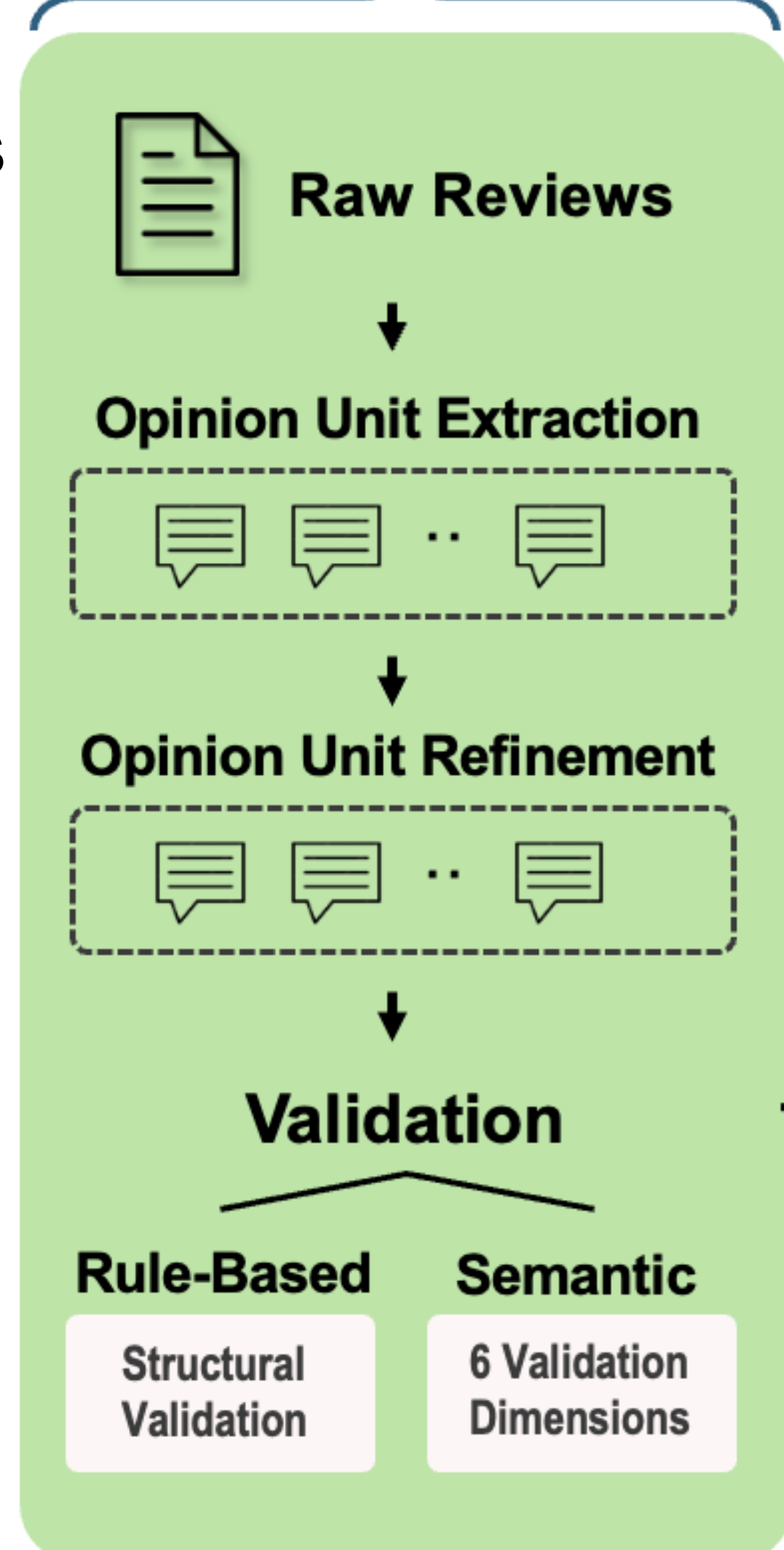
We propose an evidence-based framework, **ECRA**, that structures course reviews into validated opinion units, organizes evidence into structured evidence sets, and generates hierarchical summaries.

Methodology

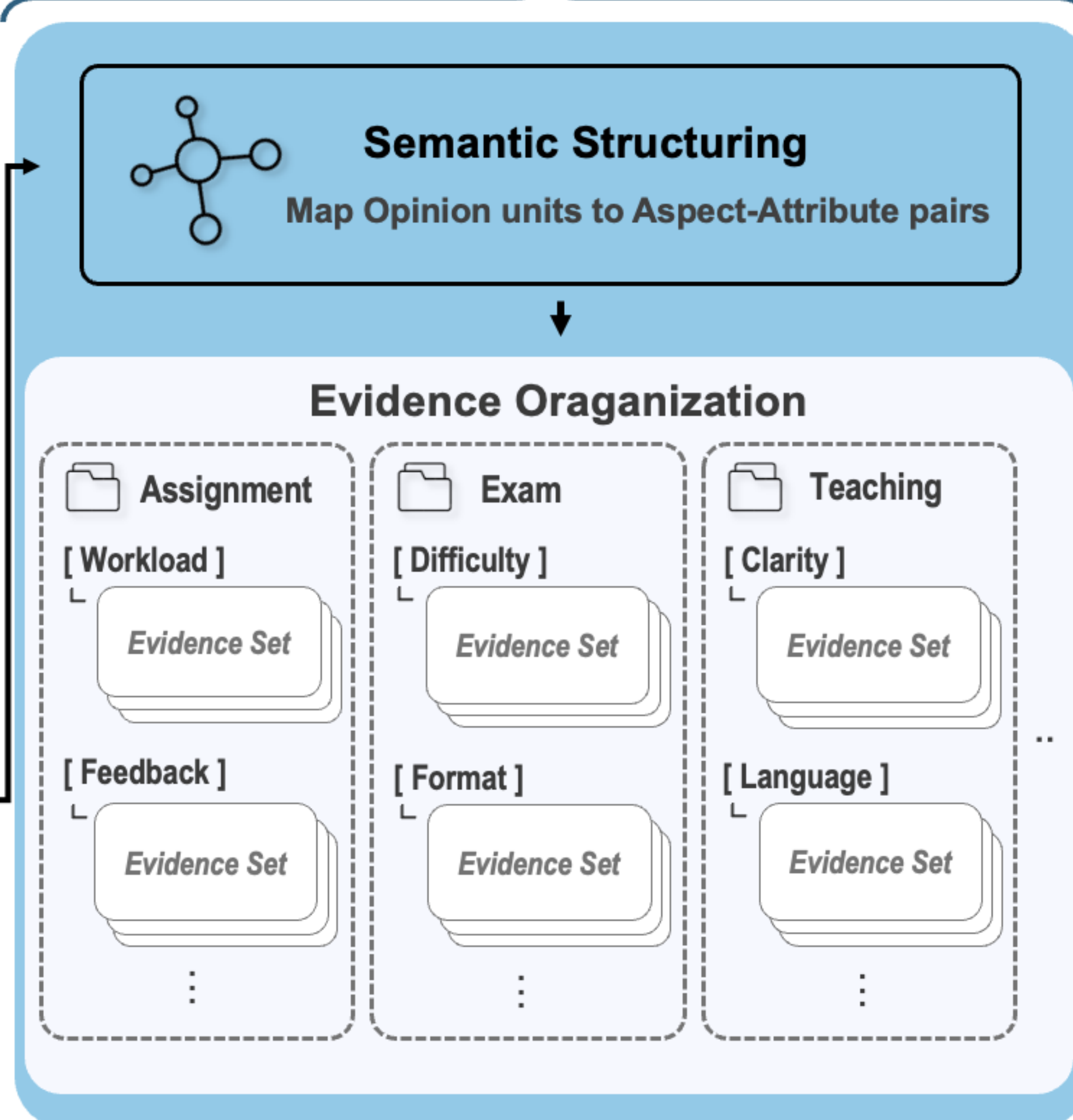
ECRA Framework

- Opinion Unit Guidelines** 📄 : A calibration-based guideline defines **atomic opinion extraction and refinement**, with an emphasis on preserving information from the original reviews
- Validation Layer** ✅ : Extracted units undergo **rule-based structural checks and semantic validation** before downstream evidence structuring
- Educational Taxonomy** 📁 : Validated units are mapped to an **Aspect-Attribute taxonomy** and organized into evidence sets for hierarchical summarization

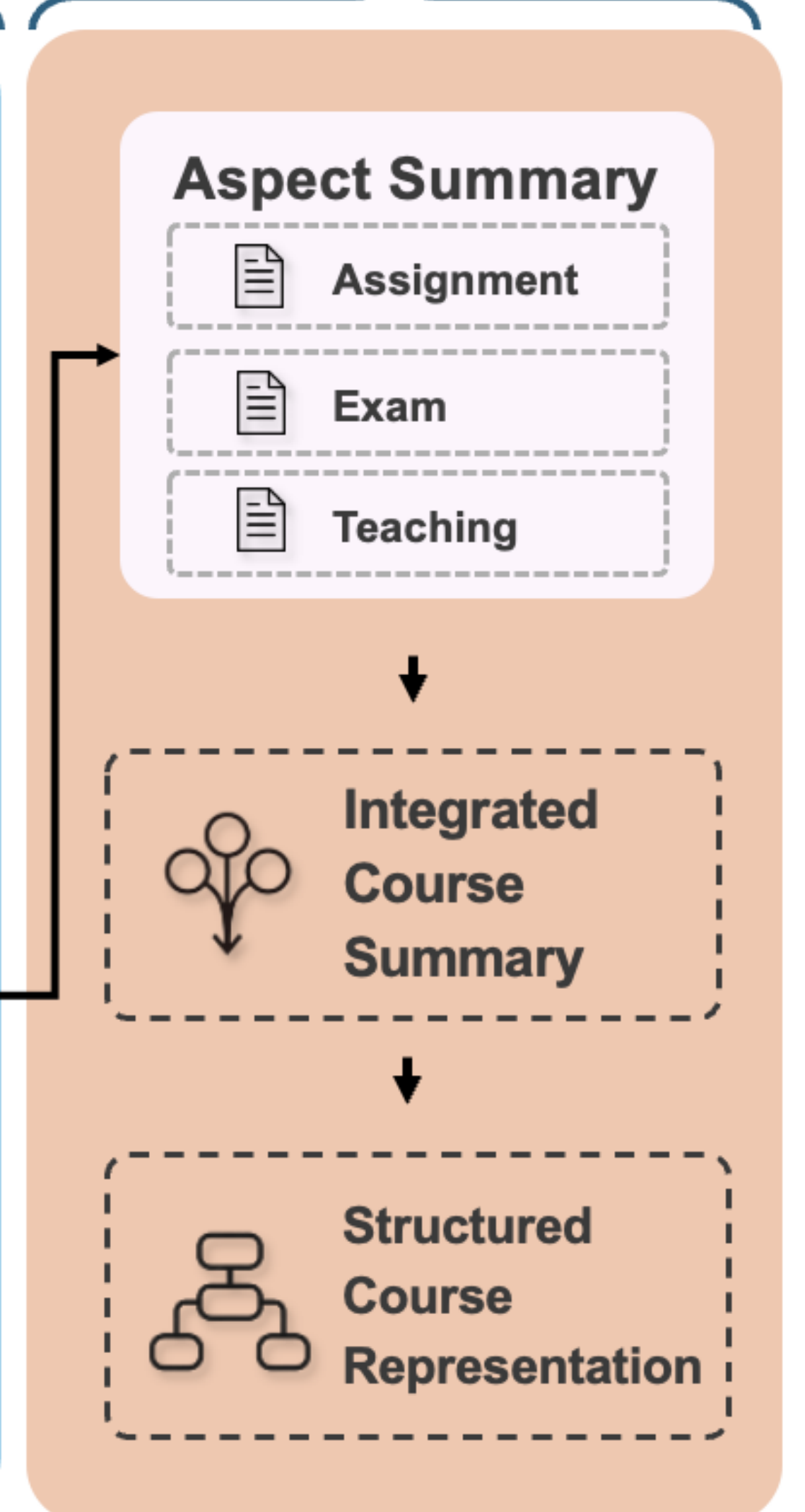
① Review Structuring



② Evidence Structuring



③ Hierarchical Summarization



Analysis

Dataset Overview

76 Courses

3,507 Course Reviews

2024–2026 Review Period

Data Sources

Course Reviews

Everytime

Course Information

2025-2 · 2026-1 Course Catalogs

Representative ECRA Summaries

Case Study1 : 선형대수

Supporting Evidence
Aspect-Attribute : 시험-난이도

- “기말 시험의 **계산량이 매우 많다.**”
- “**시험문제가 어렵지 않고** 필요한 부분만 잘 선별하여 출제된다.”

ECRA Aspect Summary

“일부 리뷰에서는 시험의 난이도가 높고 **계산량이 많다**고 평가한 반면, 다른 리뷰에서는 **시험문제가 어렵지 않다**고 언급했다.”

Case Study2 : 자료구조론

Supporting Evidence
Aspect-Attribute : 학습 자료-학습 지원 효과

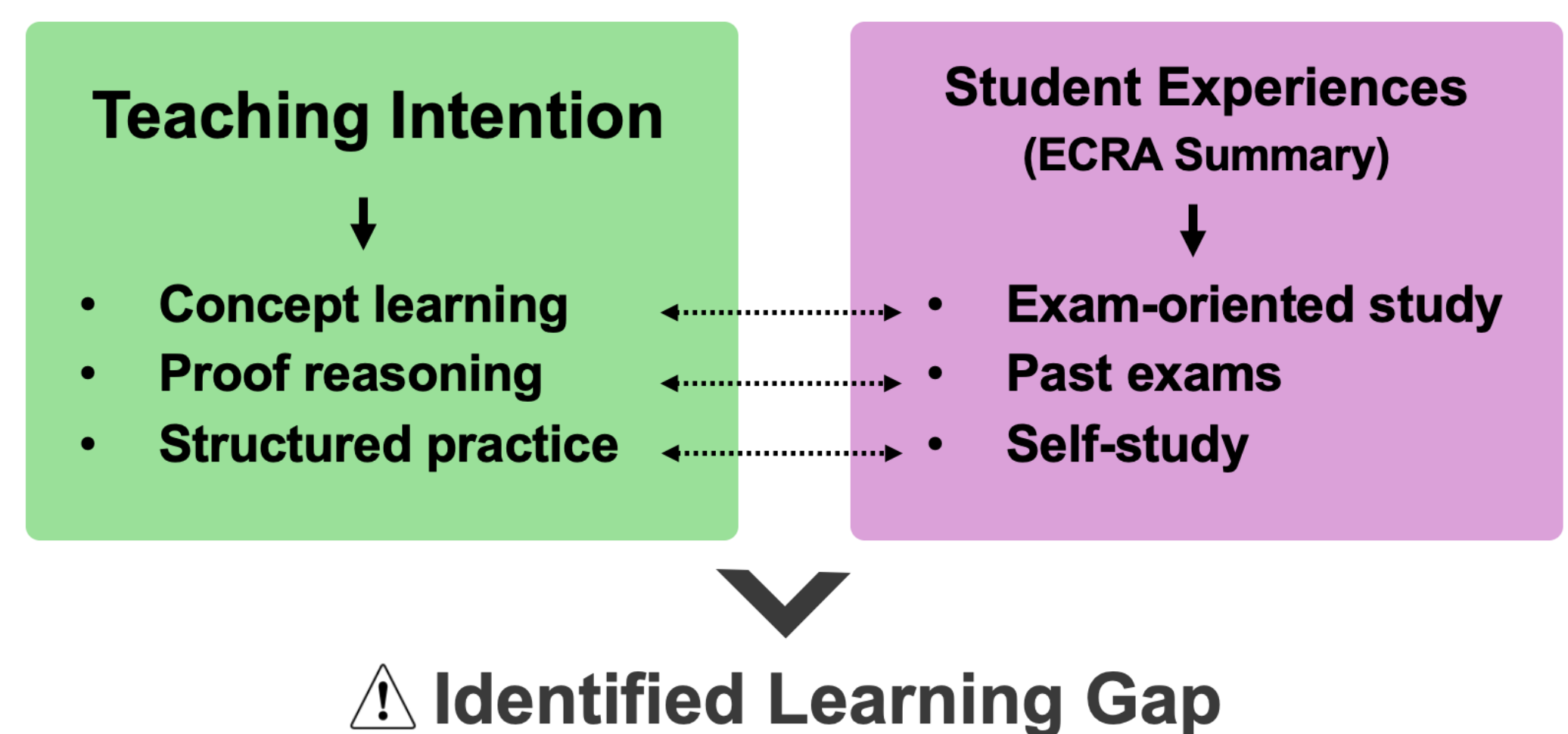
- “강의자료가 잘 구성되어 있어 **혼자 공부하기 좋다.**”
- “ PPT를 활용해 공부하면 자료구조론을 학습하는 데 도움이 된다. ”
- “강의를 듣지 않고 **PPT만으로도 공부할 수 있었다.**”

ECRA Aspect Summary

“강의자료가 잘 구성되어 있어 **혼자 공부하기 좋으며**, 일부 리뷰에서는 강의를 듣지 않고 **자료만으로도 공부할 수 있었다**는 의견이 나타났다.”

Representative Learning Gap Analysis:

Comparing teaching intentions with ECRA-generated student experience summaries enables learning gap analysis



Conclusion & Future Work

- Evidence-preserving Course Review Analysis:** ECRA transforms diverse student reviews into traceable evidence sets and hierarchical summaries, providing structured signals for Learning Gap analysis
- Socratic Learning Support:** We plan to connect structured learning-experience signals with instructional intentions to identify Learning Gaps and generate context-aware Socratic prompts for adaptive learning support